LAB 7: Switch Configuration, VLAN Setup and Inter-VLAN Routing

1. Objectives

- Understand the concept of Virtual Local Area Networks (VLANs) and their role in network segmentation.
- Configure VLANs on a Cisco Switch and assign specific ports to different VLANs.
- Implement Inter-VLAN Routing using the Router-on-a-Stick method to enable communication between different VLANs.

2. Theory

2.1 VLAN (Virtual Local Area Network)

A Virtual Local Area Network (VLAN) is a logical subdivision of a Layer 2 network that groups devices into the same broadcast domain regardless of their physical location. Unlike traditional networks where broadcast domains are defined by physical topology, VLANs allow administrators to segment a single physical switch into multiple isolated logical networks through software configuration.

Each VLAN operates as an independent broadcast domain. Frames generated by a device in one VLAN are not forwarded to devices in another VLAN. A Layer 3 device such as a router or multilayer switch is required to route traffic between VLANs. VLANs are widely used because they improve security by isolating sensitive traffic, reduce unnecessary broadcast traffic, and simplify network management by grouping devices logically by department or function rather than by physical location.

2.2 IEEE 802.1Q Trunking

Trunking enables a single physical link to carry traffic for multiple VLANs simultaneously. This is essential when connecting a switch to a router or to another switch, where frames from many VLANs must share the same physical cable. The IEEE 802.1Q standard handles trunking by inserting a 4-byte tag into the Ethernet frame header immediately after the source MAC address. This tag contains a 12-bit VLAN Identifier (VID) that tells the receiving device which VLAN the frame belongs to. The native VLAN on a trunk link is the only VLAN whose frames travel untagged.

2.3 Inter-VLAN Routing (Router-on-a-Stick)

Because VLANs create isolated broadcast domains, a Layer 3 device is required to forward packets between them. The Router-on-a-Stick (ROAS) method uses a single physical router interface that is connected to a trunk link on the switch. The physical interface is divided into

multiple logical sub-interfaces, for example Gig0/0.10 and Gig0/0.20. Each sub-interface is configured with 802.1Q encapsulation for a specific VLAN and is assigned an IP address that acts as the default gateway for all hosts in that VLAN. When a host in VLAN 10 sends a packet to a host in VLAN 20, it forwards the packet to the router sub-interface, which routes it to the correct destination VLAN.

3. Network Topology

The topology for this lab uses a Cisco 1941 Router (Router0) connected via a trunk link to a Cisco 2950-24 Switch named Sushant. Four PCs are connected to the switch and split across two VLANs as follows:

- VLAN 10 (Name: computer) — PC0 assigned 192.168.10.2 and PC1 assigned 192.168.10.3
- VLAN 20 (Name: Electronics) — PC2 assigned 192.168.20.2 and PC3 assigned 192.168.20.3

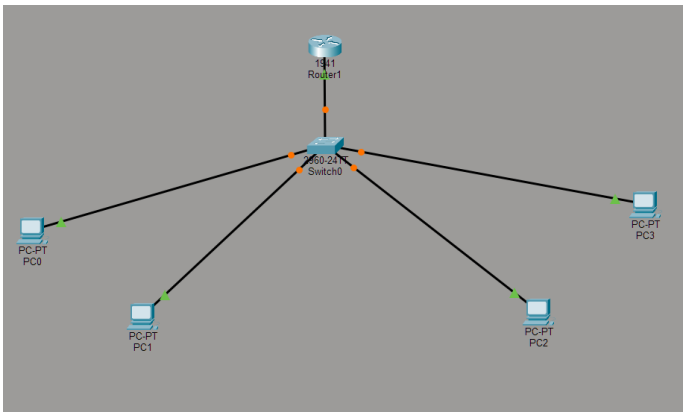
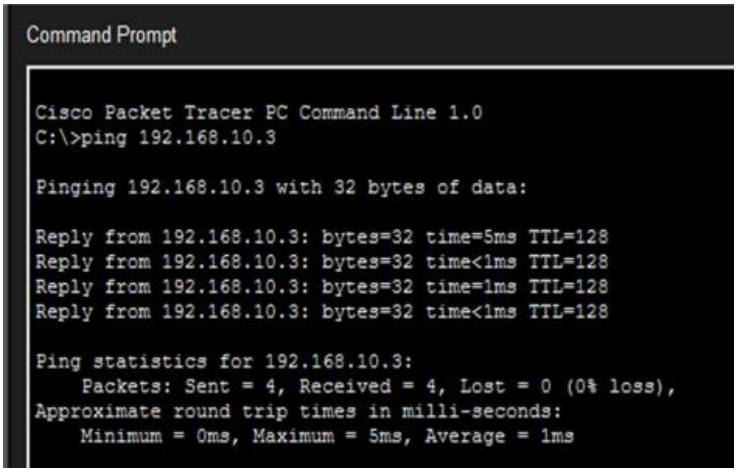


Figure 1: Network Topology — Router0 connected to Switch Sushant with four PCs across two VLANs

Testing Connection:



4. IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	VLAN
Router0	Gig0/0	N/A	N/A	N/A	N/A
Router0	Gig0/0.10	192.168.10.1	255.255.255.0	N/A	10
Router0	Gig0/0.20	192.168.20.1	255.255.255.0	N/A	20
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1	10
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1	10
PC2	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1	20
PC3	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1	20

5. Configuration

5.1 Switch Configuration — VLANs and Port Assignment

Step 1: Rename the switch and create both VLANs.

```
Switch(config)# hostname Sushant

Sushant(config)# vlan 10

Sushant(config-vlan)# name computer

Sushant(config-vlan)# exit

Sushant(config)# vlan 20

Sushant(config-vlan)# name Electronics

Sushant(config-vlan)# exit
```

Step 2: Assign access ports to their respective VLANs.

```
Sushant(config)# interface range fa0/2-3

Sushant(config-if-range)# switchport mode access

Sushant(config-if-range)# switchport access vlan 10

Sushant(config-if-range)# exit

Sushant(config)# interface range fa0/4-5

Sushant(config-if-range)# switchport mode access

Sushant(config-if-range)# switchport access vlan 20

Sushant(config-if-range)# exit
```

Step 3: Configure the uplink port to the router as a trunk.

```
Sushant(config)# interface fa0/1

Sushant(config-if)# switchport mode trunk
```

5.2 Router Configuration — Inter-VLAN Routing

Step 4: Create sub-interfaces on the router and assign 802.1Q encapsulation and an IP address to each.

```
Router0(config)# interface g0/0.10

Router0(config-subif)# encapsulation dot1q 10

Router0(config-subif)# ip address 192.168.10.1 255.255.255.0

Router0(config)# interface g0/0.20

Router0(config-subif)# encapsulation dot1q 20

Router0(config-subif)# ip address 192.168.20.1 255.255.255.0
```

6. Results and Verification

The VLANs were verified using the show vlan brief command. The output confirmed VLAN 10 (computer) with ports Fa0/2 and Fa0/3 active, and VLAN 20 (Electronics) with ports Fa0/4 and Fa0/5 active.

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
10   computer                         active    Fa0/2, Fa0/3
20   Electronics                      active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 2: Output of show VLAN brief confirming VLAN 10 and VLAN 20 on Switch Sushant

7. Discussion and Conclusion

This lab provided hands-on experience with VLAN-based network segmentation. After assigning ports to VLANs, devices in VLAN 10 and VLAN 20 were completely isolated at Layer 2, meaning they could not exchange traffic without the involvement of a router.

The Router-on-a-Stick method demonstrated how one physical router interface can serve multiple VLANs using logical sub-interfaces and 802.1Q encapsulation. Without the trunk link being properly configured on the switch, the router sub-interfaces would not receive tagged frames and inter-VLAN routing would fail entirely.

The lab was completed successfully. VLANs were created, ports were assigned correctly, trunk and sub-interface configurations were applied, and inter-VLAN communication was verified through ping tests.